

The Ego Gate

Synthesizing Psychological Valuation for Continuous AGI Learning

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Abstract

Artificial General Intelligence (AGI) systems face two existential failure modes: model collapse from synthetic data recursion, and catastrophic forgetting from unconstrained continual learning. Standard architectures lack an intrinsic mechanism to evaluate the informational value of incoming stimuli before committing to weight updates. This paper proposes the Ego Gate (V), a mathematical value filter that reverse-engineers four human psychological states — doubt, curiosity, arrogance, and dreaming — into formal, computable operations. We quantify epistemic uncertainty via predictive entropy and belief revision via a tractable, Fisher-information-weighted approximation to parameter-space KL-divergence, closely related to the natural-gradient literature [16] and to information-gain exploration bonuses in reinforcement learning [13]. The resulting signal dynamically routes incoming stimuli: discarding low-value redundancies and buffering high-value anomalies for structured experience replay. We formalize the framework, discuss its architectural implications, position it within the continual learning and intrinsic-motivation literatures, report a small-scale proof-of-concept validation on a class-incremental benchmark, and identify open research directions including adaptive coefficient learning and application to retrieval-augmented generation systems. On this benchmark, Ego-Gate-curated replay buffers reduce catastrophic forgetting by roughly a third relative to random buffer selection at the same fixed memory budget ($p < 0.001$, paired across 20 seeds) — though, as we discuss at length in Section 7, this result is preliminary and limited to a small, classification-style setting.

1. Introduction: The Stability-Plasticity Dilemma

For an artificial neural network to function as a dynamic, lifelong learner, it must receive continuous environmental stimulus. Without fresh real-world data, generative models degrade into structural collapse — a phenomenon increasingly documented as models trained on synthetic recursions of their own outputs lose distributional fidelity [1]. However, unrestricted ingestion of all incoming stimuli introduces a competing pathology: catastrophic forgetting, wherein novel parameter updates overwrite established weight pathways θ , destroying previously consolidated knowledge [2].

This tension defines the stability-plasticity dilemma: a learning system must remain plastic enough to acquire new knowledge, yet stable enough to retain prior representations. The field of continual learning has produced several approaches to this problem — Elastic Weight Consolidation (EWC) [3] penalizes updates to weights important for previous tasks; Progressive Neural Networks [4] allocate new capacity for each task; replay-based methods [5] store or generate representative samples from prior distributions to interleave with new data. Each addresses the dilemma structurally, at the

architecture or training-procedure level.

This paper proposes a complementary, pre-architectural approach: a stimulus valuation filter that operates before any weight update occurs. Rather than managing the consequences of learning indiscriminately, the Ego Gate asks a prior question: is this stimulus worth learning from at all?

The biological precedent is direct. The human brain does not consolidate every sensory input into long-term memory. An intrinsic psychological filter — loosely corresponding to what we might call ego, or self-concept — assigns value to incoming experience, discarding the mundane and consolidating the anomalous, particularly during sleep. The Ego Gate framework translates four evolved psychological mechanisms — doubt, curiosity, arrogance, and dreaming — into calculable metrics, shifting AGI architecture from static computational graphs toward biologically-inspired, dynamic value filters.

A note on scope. *The formalization that follows is built around a discrete predictive distribution $P(y_c | x_t)$, and our empirical validation (Section 5) is correspondingly a small, classification-style benchmark. We use “AGI” to describe the failure modes motivating this work and the long-term target of the research program, not as a claim that Equations (1)-(4) already operate at that scale. Sections 6.3 and 7 discuss this gap directly, and we would rather state it plainly here than let the title imply more than the formalism currently delivers.*

2. Formalization of the Ego Gate

Let x_t denote an incoming stimulus at time t . The Ego Gate computes a scalar Total Information Value $V(x_t)$ from two primary constituent functions: Doubt and Curiosity. This value is then compared against a learned threshold to determine routing behavior.

2.1 Epistemic Uncertainty — The Doubt Function

To simulate self-doubt, the model must quantify its own predictive confusion on a given stimulus. We formalize this using predictive entropy over the model’s output distribution — a measure well-established in Bayesian deep learning as a proxy for epistemic uncertainty [6].

Let $P(y_c | x_t)$ denote the probability the model assigns to outcome c given stimulus x_t . The Doubt function $D(x_t)$ is the Shannon Entropy of this distribution:

$$D(x_t) = -\sum_c P(y_c | x_t) \log P(y_c | x_t) \quad (1)$$

When the model generates a highly confident prediction, entropy approaches zero — the model experiences no doubt. A uniform distribution across outcomes maximizes entropy, flagging the stimulus as informationally valuable precisely because the model is internally conflicted. Crucially, high doubt is not equivalent to error: it signals that the stimulus occupies a region of the model’s input space where its representation is genuinely underdetermined.

A calibration caveat. Equation (1) is only a faithful measure of epistemic uncertainty to the extent that $P(y_c|x_t)$ is itself well-calibrated. Modern deep networks are frequently miscalibrated — typically overconfident — even when accurate [14], meaning raw softmax entropy can systematically understate true uncertainty in exactly the cases where the Doubt function matters most: novel or out-of-distribution stimuli. In practice we recommend pairing Equation (1) with a calibration correction (e.g. temperature scaling) or substituting an ensemble- or MC-Dropout-based uncertainty estimate [6, 9], either of which is more robust to this failure mode than a single forward pass.

2.2 Information Gain — The Curiosity Function

Curiosity is defined as the mathematical surprise generated when a new stimulus revises the model’s foundational beliefs. We measure this via Kullback-Leibler divergence between prior and posterior parameter distributions — a formulation directly motivated by the information-theoretic account of learning as Bayesian belief revision [7].

Let $P(\theta)$ represent the model’s prior parameter distribution, and $P(\theta|x_t)$ the posterior after processing x_t . The Curiosity function $C(x_t)$ is:

$$C(x_t) = D_{KL}(P(\theta|x_t) \parallel P(\theta)) = \int P(\theta|x_t) \log [P(\theta|x_t) / P(\theta)] d\theta \quad (2)$$

A stimulus perfectly aligned with existing priors yields KL-divergence of zero — the model already represents this phenomenon adequately, and no belief revision occurs. A stimulus that contradicts the model’s world-model generates large $C(x_t)$, mathematically compelling investigation and adaptation. This function formalizes the intuition that genuine curiosity is not novelty-seeking per se, but contradiction-seeking: the drive to resolve the gap between expectation and observation.

2.3 Tractable Approximation via Fisher Information

Equation (2) requires the posterior distribution over the full parameter vector θ , which is intractable for any network beyond toy scale — there is no closed-form way to represent, let alone integrate over, a posterior on millions of correlated parameters after a single observation. This is not a peripheral implementation detail; it is the central obstacle to deploying the Curiosity function in practice, and a formalization that states Equation (2) without addressing it is incomplete.

We adopt the standard remedy from information geometry. For a small parameter perturbation δ induced by a single gradient step on x_t , the KL-divergence between the resulting distribution and the original is well approximated by a quadratic form in the Fisher Information Matrix $F(\theta)$ [16]:

$$D_{KL}(P(\theta+\delta) \parallel P(\theta)) \approx \frac{1}{2} \delta^T F(\theta) \delta \quad (2')$$

where $\delta = -\eta \nabla_{\theta} L(f_{\theta}(x_t), y_t)$ is the (hypothetical) parameter update a single SGD step on x_t would induce. Taking the simplest member of this family — a diagonal, identity-scaled Fisher approximation, $F(\theta) \approx I$ — reduces Equation (2') to the squared norm of the per-example loss gradient:

$$C(x_t) \approx \|\nabla_{\theta} L(f_{\theta}(x_t), y_t)\|^2 \quad (2'')$$

This is the form we implement in Section 5. A tighter approximation would substitute the diagonal Fisher Information already maintained by EWC [3] in place of the identity, weighting each parameter’s contribution by its estimated importance — we leave this refinement, and a full variational treatment analogous to VIME [13], to future work (Section 6.3).

This connects the Curiosity function directly to the closest prior formalization of the same underlying quantity. VIME [13] also estimates an agent’s curiosity as the KL-divergence between posterior and prior beliefs about a learned dynamics model, applied as an intrinsic reward to bias exploration in reinforcement learning. VIME addresses the intractability problem above by maintaining an explicit variational posterior over a Bayesian neural network’s weights (a fully-factorized Gaussian) and computing the KL term in closed form for that family. The Ego Gate’s contribution relative to VIME is the application of an information-gain signal as an input-admission gate for continual learning — deciding what a model is permitted to learn from — rather than as a reward bonus that biases what an RL agent chooses to do. We view these as complementary uses of the same underlying information-theoretic quantity, and a tighter variational implementation of $C(x_t)$ along VIME’s lines is a natural direction for strengthening the present framework (Section 6.3).

2.4 The Value Equation and Behavioral Thresholds

The unified Information Value $V(x_t)$ is a weighted combination of Doubt and Curiosity:

$$V(x_t) = \alpha \cdot D(x_t) + \beta \cdot C(x_t) \quad (3)$$

where $\alpha, \beta \in \mathbb{R}_{\geq 0}$ are sensitivity coefficients governing the model’s relative responsiveness to uncertainty versus belief revision. When $\alpha > \beta$, the system prioritizes inputs where it is internally confused (high entropy); when $\beta > \alpha$, it prioritizes inputs that maximally update its beliefs. In the base formulation, α and β are treated as hyperparameters; Section 6.1 addresses their adaptive formulation — and Section 5 reports a preliminary empirical result bearing directly on whether the naive $\alpha=\beta=1$ choice is adequate.

Incoming stimuli are routed via the Arrogance Threshold τ :

$$\begin{aligned} \text{Action} = \{ & \text{Discard (Arrogance) if } V(x_t) < \tau \\ & \{ \text{Push to Buffer B (Dreaming) if } V(x_t) \geq \tau \end{aligned} \quad (4)$$

The threshold τ defines the boundary between the two behavioral modes formalized in Section 3, which also introduces a second threshold for a third mode.

3. Architectural Implications

3.1 Arrogance — Redundancy Filtration

When $V(x_t) < \tau$, the system acts with arrogance: the stimulus is discarded without triggering backpropagation. This has two concrete consequences. First, it provides substantial computational savings by bypassing the gradient computation pipeline for low-value inputs. Second, and more importantly, it protects foundational weight pathways from erosion by redundant data — the primary mechanism of catastrophic forgetting in naive continual learning systems.

The term arrogance is deliberately chosen: the system is not passively ignoring low-value stimuli, but actively asserting that its current representation is sufficient — that nothing in this stimulus warrants revision. This is not merely a rhetorical flourish: confidently discounting new evidence that does not update one’s current model is the operational signature of confirmation bias, among the most thoroughly documented biases in human cognition [15]. The Ego Gate’s Arrogance mode is, in effect, a deliberately engineered version of this same mechanism, adopted here for its computational benefits rather than treated purely as a flaw to correct — though we note in Section 7 that this grounding is best read as inspiration for the label, not a claim that human confirmation bias and this thresholding rule are the same mechanism.

3.2 Dreaming — Structured Experience Replay

When $V(x_t) \geq \tau$, the stimulus is placed into an active memory buffer B . During designated offline computation periods — the Dreaming phase — the model performs structured experience replay: buffered high-value stimuli are synthesized alongside sampled historical core data and used to update weights in a controlled, interleaved manner.

This approach is directly motivated by replay-based continual learning methods [5] and the complementary learning systems theory of biological memory consolidation [8], which proposes that the hippocampus rapidly encodes new experiences while the neocortex slowly integrates them during sleep. The Ego Gate formalizes the selection criterion for what enters this replay buffer — a step that existing replay methods typically handle via random sampling or task boundaries, rather than principled information-theoretic valuation. Section 5 reports a first empirical test of whether that principled valuation actually outperforms random sampling at a fixed buffer size.

Formally, the replay objective during the Dreaming phase minimizes:

$$L_{\text{dream}} = E_{x \sim B} [L(f_{\theta}, x)] + \lambda \cdot E_{x \sim P_{\text{core}}} [L(f_{\theta}, x)] \quad (5)$$

where P_{core} represents the distribution of historical core training data, λ is a stability coefficient, and the joint optimization prevents both forgetting and overfitting to the buffered anomalies.

3.3 Active Learning — Human Oracle Intervention

A third behavioral mode emerges when the Doubt function alone approaches its theoretical maximum — near-uniform predictive entropy indicating complete model confusion. In this regime, the system enters a quarantine state: parameter updates are paused and human oracle intervention is requested. This provides a principled mechanism for detecting potential data poisoning or out-of-distribution adversarial inputs, and creates a human-in-the-loop checkpoint at the boundary of the model’s competence.

We formalize this with a second, higher threshold τ' applied to $D(x_t)$ alone, extending the routing rule of Equation (4):

$$\begin{aligned} \text{Action} = & \{ \text{Discard (Arrogance)} \text{ if } V < \tau \\ & \{ \text{Push to Buffer B (Dreaming)} \text{ if } V \geq \tau \text{ and } D < \tau' \\ & \{ \text{Quarantine (Active Learning)} \text{ if } D \geq \tau' \end{aligned} \quad (4')$$

τ' is set independently of τ because quarantine is triggered by confusion in isolation — a stimulus can produce high Curiosity (large belief revision) without approaching maximal entropy, and such stimuli are precisely the ones we want the system to learn from autonomously (Dreaming), not refer to a human. Quarantine is reserved for the regime where the model’s predictive distribution is so close to uniform that no class is preferred, which is the operating definition we use for “complete model confusion” and a plausible signature of data poisoning or severe distribution shift.

4. Positioning Within Continual Learning Literature

The Ego Gate framework synthesizes ideas from several established research threads while proposing a novel unification.

Continual learning and catastrophic forgetting. The core challenge this framework addresses is well-documented [2, 3, 4, 5]. EWC [3] protects important weights via a quadratic penalty; the Ego Gate protects them via input filtration before gradient computation. These approaches are orthogonal and potentially composable: the Ego Gate determines which inputs are processed, while EWC determines which weights are updated.

Bayesian deep learning and uncertainty quantification. The Doubt function (Eq. 1) directly employs predictive entropy, a standard uncertainty measure in Bayesian neural networks and Monte Carlo Dropout approaches [6, 9]. The Curiosity function (Eq. 2) draws on variational inference and information-theoretic accounts of learning [7].

Intrinsic motivation in reinforcement learning. The Curiosity function (Eq. 2) is closely related to — and was independently motivated by the same information-theoretic intuition as — Variational Information Maximizing Exploration (VIME) [13], which uses the KL-divergence between posterior and prior beliefs over a learned dynamics model as an intrinsic reward for exploration. This is a closer formal match than the count-based [10] or prediction-error-based [11] approaches, which use different (non-KL) novelty

signals. The distinction from VIME is one of application, not formulation: VIME biases which actions an RL agent takes, by adding the information-gain term to the external reward; the Ego Gate instead uses the same kind of term to decide whether a supervised or self-supervised model is permitted to update on a given stimulus at all, independent of any reward signal. We view the Ego Gate’s Curiosity function as VIME’s information-gain criterion repurposed from an exploration bonus into an input-admission gate — see Section 2.3 for the tractable approximation we use in place of VIME’s full variational treatment.

Replay-based continual learning. Experience replay [5] and generative replay [12] methods are the closest architectural antecedents to the Dreaming phase. The Ego Gate’s contribution is providing a principled, information-theoretic criterion for buffer population — replacing random or task-boundary-based selection with stimulus valuation. Section 5 tests this claim directly.

5. Empirical Validation: A Class-Incremental Proof of Concept

The framework as originally presented was purely theoretical. This section reports a small-scale proof-of-concept experiment testing its central, falsifiable claim: that curating a bounded replay buffer by Information Value $V(x_t)$ reduces catastrophic forgetting more than curating it by uniform random sampling, at the same fixed memory budget.

5.1 Setup

Data and task. We use scikit-learn’s *digits* dataset (1,797 8×8 grayscale images of handwritten digits 0-9, 64 pixel features normalized to $[0,1]$). We construct a standard class-incremental split: Task A = digits $\{0,1,2,3,4\}$ (901 images), Task B = digits $\{5,6,7,8,9\}$ (896 images), each split 80/20 into train/test.

Model. A single-hidden-layer MLP ($64 \rightarrow 64$ ReLU $\rightarrow 10$ softmax, $\approx 4,800$ parameters), trained with plain mini-batch SGD (no momentum or Adam, so per-example gradients used for scoring are transparent and easy to verify).

Procedure. (1) Train to convergence on Task A (60 epochs, $lr=0.5$, $batch=32$); this checkpoint is the shared starting point for every condition below. (2) Score every Task A training example using the frozen checkpoint: Doubt via Eq. (1); Curiosity via the practical approximation of Eq. (2’’) ($F \approx I$, the simplest member of the family in Section 2.3). (3) Combine via $V = z\text{-score}(\text{Doubt}) + z\text{-score}(\text{Curiosity})$, i.e. $\alpha=\beta=1$, the naive unweighted baseline. (4) Build six replay buffers of at most $K=40$ examples ($\approx 5.6\%$ of the 720 Task-A training examples): **None** (empty), **Random** (uniform sample), **EgoGate** (top- K by V), **DoubtOnly** (top- K by Doubt alone), **CuriosityOnly** (top- K by Curiosity alone), and **Full** (all 720 examples, an unbounded oracle upper bound). (5) From the same checkpoint, continue training for 40 epochs on Task-B data interleaved with the buffer (each condition starts from a fresh copy of the checkpoint). (6) Evaluate accuracy on the held-out Task A test set (retention) and Task B test set (plasticity). (7) Repeat the full pipeline — fresh initialization, fresh random buffer draw — for 20 seeds; report mean $\pm$ standard deviation, with paired t-tests across seeds.

5.2 Results

Condition	Task A acc. (pre-Task B)	Task A acc. (post-Task B)	Forgetting	Task B acc.
None	0.994	0.004 ± 0.012	0.991 ± 0.012	0.998 ± 0.003
Random	0.994	0.820 ± 0.043	0.174 ± 0.043	0.994 ± 0.004
EgoGate	0.994	0.877 ± 0.033	0.118 ± 0.032	0.992 ± 0.007
DoubtOnly	0.994	0.883 ± 0.036	0.111 ± 0.035	0.992 ± 0.005
CuriosityOnly	0.994	0.854 ± 0.036	0.140 ± 0.035	0.995 ± 0.003
Full (oracle)	0.994	0.981 ± 0.006	0.013 ± 0.006	0.981 ± 0.005

Table 1. Task A retention and Task B plasticity after continual training, by buffer-selection condition ($K=40$, except None and Full). Mean $\pm$ std over 20 seeds.

Significance. Paired t-tests on forgetting across the same 20 seeds: Random vs. EgoGate, $t=4.22$, $p=0.0005$; Random vs. DoubtOnly, $t=5.42$, $p<0.0001$; Random vs. CuriosityOnly, $t=3.17$, $p=0.0051$; Random vs. Full, $t=15.11$, $p<0.0001$. EgoGate vs. DoubtOnly: $t=0.74$, $p=0.47$ (not significant).

Without replay, the network exhibits near-total catastrophic forgetting (99.1 points of Task A accuracy lost), confirming the benchmark induces the failure mode the framework targets. All three information-theoretic selection criteria — EgoGate, DoubtOnly, and CuriosityOnly — significantly reduce forgetting relative to random selection at the same fixed buffer size ($p \leq 0.005$ for all three), supporting the central hypothesis of this section: informative buffer curation outperforms random retention under a constrained memory budget.

A more interesting, and humbler, finding: Doubt alone was the strongest individual predictor of buffer value in this setting, and the unweighted Doubt+Curiosity combination (EgoGate) was statistically indistinguishable from Doubt alone ($p=0.47$) — it did not significantly improve on Doubt, and numerically trailed it slightly. This is an informative negative result rather than a failure: it suggests the naive $\alpha=\beta=1$ weighting is not optimal here, and lends direct empirical motivation to the adaptive-coefficient-learning direction in Section 6.1, rather than supporting an implicit assumption that equal weighting is sufficient. The Full-replay oracle nearly eliminates forgetting (1.3% residual), confirming the upper bound and indicating that the gap between EgoGate and Full is attributable to the bounded buffer size K , not a flaw in the underlying task.

5.3 Limitations of This Evaluation

This experiment uses one small dataset, one architecture, one task boundary, and one fixed combination rule; Section 7 discusses how far these results can and cannot be generalized. We report it as a first, falsifiable test of the framework’s central claim — not as evidence that the Ego Gate is validated at the scale the Introduction motivates it for.

6. Open Research Directions

6.1 Adaptive Coefficient Learning

The sensitivity coefficients α and β in Equation (3) govern the system’s relative weighting of uncertainty versus information gain. In the base formulation, these are fixed hyperparameters. A natural extension is to learn them adaptively: treating α and β as outputs of a meta-controller trained via reinforcement learning, where rewards reflect downstream task performance and memory stability. The threshold τ itself admits similar treatment.

We note that meta-RL controllers of this kind are themselves notoriously sample-inefficient and can introduce their own training instability — using a hard problem (gating credit assignment) to solve another hard problem (online policy learning) is not obviously a net simplification. A more immediately tractable interim approach, and the one we would prioritize first, is periodic grid search or Bayesian optimization over (α, β, τ) against a small held-out validation stream, re-fit on a fixed schedule (e.g. at each task boundary) rather than online. Our own result in Section 5 — that the equal-weighted combination underperformed Doubt alone — suggests that even this simpler, periodic re-weighting would likely yield a meaningful improvement before resorting to a learned meta-controller.

6.2 Application to Retrieval-Augmented Generation

A promising near-term application is adapting the Ego Gate as a response-depth calibration layer in retrieval-augmented generation (RAG) systems. Current answer engines treat each query as an isolated retrieval task, with no model of the user’s knowledge state or the conceptual dependencies between topics. The Doubt function could measure the system’s uncertainty about user comprehension rather than output class probability; the Curiosity function could measure how far the current answer deviates from what a user at this knowledge level actually requires. The Ego Gate would then determine whether to retrieve deeper, restructure the explanation, or surface a missing conceptual prerequisite.

6.3 Scaling Beyond Toy Benchmarks

Section 5 provides a first proof of concept on a small, two-task, classification-style benchmark. The natural next steps are: (1) repeating the evaluation on standard continual-learning benchmarks with more tasks and higher-dimensional inputs (Split-CIFAR, Permuted-MNIST, CORe50); (2) replacing the identity-scaled Fisher approximation in Section 2.3 with the diagonal Fisher already computed by EWC, or with a full variational treatment along the lines of VIME [13]; (3) testing whether the qualitative finding that Doubt alone matches or exceeds the naive Doubt+Curiosity combination holds at larger scale, or whether Curiosity’s relative contribution grows with task diversity; and (4) extending the formalism beyond fixed-class classification to token-level or embedding-level stimuli, which is the necessary bridge between the present framework and the generative, foundation-model settings invoked in the Introduction.

7. Limitations and Scope

We are explicit here about what this paper does and does not establish, consolidating points raised throughout.

Scope of the formalism. Equations (1)–(4) are defined over a discrete output distribution $P(y_c|x_t)$, the natural setting for classification-style continual learners. The framework as written does not directly specify how Doubt and Curiosity would be computed for token-level generative models, where “the prediction” is a distribution over a large vocabulary at every position rather than a single distribution over a small fixed label set. Bridging this gap — most plausibly via the RAG application sketched in Section 6.2, or via a sequence-level reformulation of Equations (1)–(2) — is the most important piece of unfinished work in this line of research.

Scope of the empirical validation. The result in Section 5 uses one small dataset (1,797 8×8 images), one architecture (a single-hidden-layer MLP), one task boundary, and one non-adaptive, equally-weighted combination rule. It is evidence that information-theoretic buffer curation can outperform random curation at a fixed memory budget in at least one small setting — consistent with, but far short of confirming, the broader claims in the Introduction. We would caution against citing this result as evidence that the Ego Gate “works” for AGI-scale continual learning; it is evidence that the core mechanism is not obviously wrong, which is a substantially weaker claim.

Scope of the Curiosity approximation. The $C(x_t)$ used in Section 5 sets $F(\theta) \approx I$ in Equation (2’), discarding the per-parameter importance weighting that makes the Fisher-information approximation more than a generic gradient-norm heuristic. The reported results should be read as a lower bound on what a properly Fisher-weighted or fully variational (VIME-style) implementation of Curiosity could achieve.

Psychological grounding. We use doubt, curiosity, arrogance, and dreaming as organizing labels for formal operations that have genuine, independently-motivated grounding in Bayesian uncertainty estimation, information-theoretic belief revision,

confirmation bias, and complementary learning systems theory, respectively [6, 7, 8, 15]. We do not claim that these four labels constitute an exhaustive or validated taxonomy of the psychological mechanisms underlying human memory consolidation; the mapping is best read as inspiration and exposition rather than a tested theory of human cognition.

8. Conclusion

The transition from narrow AI to general intelligence requires systems that do not merely process data, but actively evaluate its worth. The Ego Gate provides a mathematical framework for continuous learning models to self-regulate cognitive load — discarding the redundant, buffering the anomalous, and protecting foundational knowledge from both noise and forgetting.

The framework’s four psychological primitives — doubt as predictive entropy, curiosity as KL-divergence (here approximated via Fisher information, Section 2.3), arrogance as principled discarding grounded in the mechanics of confirmation bias, and dreaming as structured replay — are not merely evocative labels. Each is tied to a formal operation and, in the case of doubt and curiosity combined, to a measurable reduction in catastrophic forgetting in a small proof-of-concept benchmark (Section 5). Taken together, they constitute a pre-architectural selectivity layer largely absent from current continual learning approaches — though substantial work remains: scaling beyond toy benchmarks, tightening the Curiosity approximation, and learning the $\alpha/\beta/\tau$ coefficients adaptively, before this claim can be made at the scale the title invokes. Whether implemented as a standalone filter or integrated into existing EWC or replay frameworks, the Ego Gate proposes that the path to stable, lifelong learning runs through a question every intelligent system must eventually answer: what is this worth knowing?

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